Akoizo Original Practice - Form 1 - Answer key

Division B | Anatomy And Physiology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) The sternum is ___ to the heart and the elbow is ___ to the wrist.

A. anterior; proximal

B. posterior; distal

C. anterior; distal

D. superior; proximal

Correct: A

2. (2 points) Aspirated peanut most likely lodges in which bronchus and why?

Unit: Airways: From Nose to Bronchioles.
Worked-case data: Tube A has the baseline radius. Tube B has 80% of that radius. Both have the same length, viscosity, and driving pressure.

A. Right, wider and more vertical

B. Left, more horizontal

C. Trachea only

D. Left, narrower

Correct: A

3. (2 points) Vital capacity equals:

Unit: Ventilation, Volumes, and Spirometry.
Worked-case data: Person A breathes 500 mL twelve times each minute. Person B breathes 250 mL twenty-four times. Both have 150 mL anatomical dead space per breath.

A. ERV + RV

B. TV + RV

C. TV + IRV

D. IRV + TV + ERV

Correct: D

4. (2 points) Reversible obstruction with eosinophils best fits:

Unit: Obstructive Disease: Asthma, Bronchitis, Emphysema, COPD.
Worked-case data: Dataset A has variable wheeze, reduced expiratory ratio, and largely preserved diffusion capacity. Dataset B has persistent obstruction, increased residual volume, and reduced diffusion capacity.

A. Pneumothorax

B. Emphysema

C. Asthma

D. Fibrosis

Correct: C

5. (2 points) Pepsinogen is converted to pepsin by:

Unit: Alimentary Canal: Mouth to Stomach.
Worked-case data: An educational case describes loss of parietal-cell function with adequate dietary B12 but poor B12 absorption.
Separate investigation observations:
Diet record: The case provides adequate vitamin B12 intake.
Gastric finding: Parietal-cell function is greatly reduced.
Distal finding: The terminal ileum is anatomically present, but normal B12 uptake is impaired.

A. Bile

B. HCl/acid pH

C. Bicarbonate

D. Insulin

Correct: B

6. (2 points) Duodenal ulcer pain classically:

Unit: Ulcers, GI Bleeding, and GI Cancers.
Worked-case data: Case A supplies impaired mucus protection and a focal mucosal defect. Case B supplies an invasive colonic lesion that narrows the lumen.
Separate investigation observations:
Mucosal image description: A focal defect extends through the mucosa.
Protective function: Mucus-bicarbonate protection is reduced.
Lumen assessment: No obstructing tissue mass is described.

A. Better with food/antacid

B. No relation

C. Only nocturnal

D. Worse with food

Correct: A

7. (2 points) B cells mature in the ___; T cells mature in the ___.

Unit: Central Immune Organs and Lymph Flow.
Worked-case data: A small skin abrasion introduces material into tissue fluid of the left foot. Trace a plausible route toward immune surveillance and return to blood.
Separate investigation observations:
Origin: Antigen enters interstitial fluid around a skin abrasion.
Transport: Afferent lymph carries antigen-bearing cells from the tissue.
Cell encounter: Mature naive lymphocytes meet presented antigen in organized lymphoid tissue.

A. Spleen; node

B. Node; spleen

C. Thymus; marrow

D. Marrow; thymus

Correct: D

8. (2 points) MAC is formed by:

Unit: Innate Immunity and Complement.
Worked-case data: A hypothetical dataset shows poor complement-mediated coating of microbes, while antibody production is present. A second dataset shows coating but failure of terminal membrane attack.
Separate investigation observations:
Recognition assay: Microbial targets are recognized.
Coating assay: C3b deposition on targets is markedly reduced.
Phagocyte assay: Phagocytes work better when an external opsonin is supplied.

A. Histamine

B. C3b only

C. C5b-9

D. IgE alone

Correct: C

9. (2 points) AIDS is defined by CD4 under ___ or opportunistic illness.

A. 500

B. 350

C. 200

D. 50

Correct: C

10. (2 points) Which cut through a straight segment of intestine best shows concentric wall layers?

A. Perpendicular to the segment’s long axis

B. Always midsagittal

C. Always frontal

D. Parallel to its long axis

Correct: A

11. (4 points) Which cells and structures drive mucociliary clearance?

Unit: Airways: From Nose to Bronchioles.
Worked-case data: Tube A has the baseline radius. Tube B has 80% of that radius. Both have the same length, viscosity, and driving pressure.

answer: Ciliated pseudostratified cells plus goblet mucus; cilia beat toward pharynx. (4 points).

12. (2 points) FEV1/FVC 0.55 with reduced FEV1 suggests:

A. Mixed only

B. Restriction

C. Obstruction

D. Normal

Correct: C

13. (2 points) Low DLCO with bullae points to:

Unit: Obstructive Disease: Asthma, Bronchitis, Emphysema, COPD.
Worked-case data: Dataset A has variable wheeze, reduced expiratory ratio, and largely preserved diffusion capacity. Dataset B has persistent obstruction, increased residual volume, and reduced diffusion capacity.

A. Asthma

B. Chronic bronchitis

C. Emphysema

D. Pneumonia

Correct: C

14. (2 points) Intrinsic factor is made by ___ and is needed for:

Unit: Alimentary Canal: Mouth to Stomach.
Worked-case data: An educational case describes loss of parietal-cell function with adequate dietary B12 but poor B12 absorption.
Separate investigation observations:
Diet record: The case provides adequate vitamin B12 intake.
Gastric finding: Parietal-cell function is greatly reduced.
Distal finding: The terminal ileum is anatomically present, but normal B12 uptake is impaired.

A. Parietal cells; B12 absorption

B. Chief cells; lipase

C. Goblet cells; mucus

D. G cells; gastrin

Correct: A

15. (2 points) Best H. pylori confirmation after treatment:

Unit: Ulcers, GI Bleeding, and GI Cancers.
Worked-case data: Case A supplies impaired mucus protection and a focal mucosal defect. Case B supplies an invasive colonic lesion that narrows the lumen.
Separate investigation observations:
Mucosal image description: A focal defect extends through the mucosa.
Protective function: Mucus-bicarbonate protection is reduced.
Lumen assessment: No obstructing tissue mass is described.

A. Serum IgG alone

B. No testing

C. Symptoms only

D. Urea breath/stool antigen

Correct: D

16. (2 points) Asplenic patients need vaccines against:

Unit: Central Immune Organs and Lymph Flow.
Worked-case data: A small skin abrasion introduces material into tissue fluid of the left foot. Trace a plausible route toward immune surveillance and return to blood.
Separate investigation observations:
Origin: Antigen enters interstitial fluid around a skin abrasion.
Transport: Afferent lymph carries antigen-bearing cells from the tissue.
Cell encounter: Mature naive lymphocytes meet presented antigen in organized lymphoid tissue.

A. No vaccines

B. Viruses only

C. Encapsulated bacteria

D. Fungi only

Correct: C

17. (2 points) C3b mainly functions to:

A. Vasoconstrict

B. Opsonize for phagocytosis

C. Block fever

D. Make IgE

Correct: B

18. (2 points) Why can a helper T-cell defect affect several immune pathways?

Unit: Immunodeficiency: HIV, SCID, and CVID.
Worked-case data: A hypothetical case has circulating B cells but repeatedly poor antibody responses to supplied antigen challenges. T-cell measurements are also provided and appear preserved.
Separate investigation observations:
Cell inventory: B cells are present in the sample.
Challenge response: Antibody output after repeated specified challenges is poor.
Comparison: The supplied T-cell functional comparison is preserved.

A. T cells are all antibodies

B. Only red cells are affected

C. All immune cells have identical jobs

D. Helper signals coordinate multiple responses

Correct: D